

15.1 Solution: (a) From Eq. (14.65), we have

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{n} \times [(\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]}{(1 - \mathbf{n} \cdot \boldsymbol{\beta})^2} e^{i\omega(t - \mathbf{n} \cdot \mathbf{r}/c)} \\ &= \left\{ \mathbf{n} \times [(\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}] \right\} (1 + 2\mathbf{n} \cdot \boldsymbol{\beta})(1 - i k \mathbf{n} \cdot \mathbf{r}) e^{i\omega t}.\end{aligned}$$

The first term can be written as

$$\begin{aligned}\mathbf{n} \times [(\mathbf{n} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}] &= \mathbf{n} \times (\mathbf{n} \times \dot{\boldsymbol{\beta}}) - \mathbf{n} \times (\boldsymbol{\beta} \times \dot{\boldsymbol{\beta}}) \\ &= -\dot{\boldsymbol{\beta}}_{\perp} - \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta} + \beta_{\parallel} \dot{\boldsymbol{\beta}},\end{aligned}$$

where we have defined, for a vector $\mathbf{a}$, $a_{\parallel} = \mathbf{n} \cdot \mathbf{a}$ and $a_{\perp} = (\mathbf{n} \times \mathbf{a}) \times \mathbf{n}$. Then,

$$\begin{aligned}e^{-i\omega t} \mathbf{I} &= -(\dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta} - \beta_{\parallel} \dot{\boldsymbol{\beta}})(1 + 2\beta_{\parallel})(1 - i k r_{\parallel}) \\ &= -(1 + 2\beta_{\parallel} - i k r_{\parallel})(\dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta} - \beta_{\parallel} \dot{\boldsymbol{\beta}}) \\ &= -(\dot{\boldsymbol{\beta}}_{\perp} - i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + 2\beta_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta} - \beta_{\parallel} \dot{\boldsymbol{\beta}}) \\ &= -(\dot{\boldsymbol{\beta}}_{\perp} - i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + 2\beta_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta}_{\parallel} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta}_{\perp} - \beta_{\parallel} \dot{\boldsymbol{\beta}}_{\parallel} - \beta_{\parallel} \dot{\boldsymbol{\beta}}_{\perp}) \\ &= -(\dot{\boldsymbol{\beta}}_{\perp} - i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + \beta_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta}_{\perp}),\end{aligned}$$

where we have used the fact that

$$\dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta}_{\parallel} = (\mathbf{n} \cdot \dot{\boldsymbol{\beta}})(\mathbf{n} \cdot \boldsymbol{\beta}) \mathbf{n} = \beta_{\parallel} \dot{\boldsymbol{\beta}}_{\parallel}.$$

By a difference in sign, we have

$$\begin{aligned}\mathbf{J} &= -e \int \mathbf{I} dt \\ &= e \int dt e^{i\omega t} (\dot{\boldsymbol{\beta}}_{\perp} + \beta_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} + \dot{\boldsymbol{\beta}}_{\parallel} \boldsymbol{\beta}_{\perp} - i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp}) \\ &= e \int dt e^{i\omega t} \left(\dot{\boldsymbol{\beta}}_{\perp} + \frac{d}{dt} (\beta_{\parallel} \boldsymbol{\beta}_{\perp}) - i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} \right).\end{aligned}$$

(b) First, by an integration by parts, we can show that

$$-\int i k r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} e^{i\omega t} dt = -\frac{1}{c} \int r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp} d(e^{i\omega t}) = \frac{1}{c} \int \frac{d}{dt} (r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp}) e^{i\omega t} dt,$$

with proper convergence factor taken into account. Then,

$$\begin{aligned}\mathbf{J} &= e \int \left(\dot{\boldsymbol{\beta}}_{\perp} + \frac{d}{dt} (\beta_{\parallel} \boldsymbol{\beta}_{\perp}) + \frac{1}{c} \frac{d}{dt} (r_{\parallel} \dot{\boldsymbol{\beta}}_{\perp}) \right) e^{i\omega t} dt \\ &= e \int \left(\dot{\boldsymbol{\beta}}_{\perp} + \frac{1}{c} \frac{d}{dt} \left[\frac{d}{dt} (\mathbf{n} \cdot \mathbf{r}) \boldsymbol{\beta}_{\perp} + (\mathbf{n} \cdot \mathbf{r}) \frac{d\boldsymbol{\beta}_{\perp}}{dt} \right] \right) e^{i\omega t} dt \\ &= e \int \left(\dot{\boldsymbol{\beta}}_{\perp} + \frac{1}{c} \frac{d^2}{dt^2} [(\mathbf{n} \cdot \mathbf{r}) \boldsymbol{\beta}_{\perp}] \right) e^{i\omega t} dt.\end{aligned}$$

Since

$$(\mathbf{n} \cdot \mathbf{r}) \boldsymbol{\beta}_{\perp} = \frac{1}{2} [(\mathbf{n} \cdot \mathbf{r}) \boldsymbol{\beta}_{\perp} + (\mathbf{n} \cdot \boldsymbol{\beta}_{\perp}) \mathbf{r}] + \frac{1}{2} (\mathbf{r} \times \boldsymbol{\beta}_{\perp}) \times \mathbf{n} = \frac{1}{2} (\mathbf{n} \cdot \mathbf{r}) \boldsymbol{\beta}_{\perp} + \frac{1}{2} (\mathbf{r} \times \boldsymbol{\beta}_{\perp}) \times \mathbf{n}$$

$$= \frac{1}{2}(\mathbf{r} \times \boldsymbol{\beta}) \times \mathbf{n} - \frac{1}{2}(\mathbf{r} \times \boldsymbol{\beta}_{\parallel}) \times \mathbf{n} + \frac{1}{2}(\mathbf{n} \cdot \mathbf{r})\boldsymbol{\beta}_{\perp},$$

as $\mathbf{n} \cdot \boldsymbol{\beta}_{\perp} = 0$ and $\boldsymbol{\beta} = \boldsymbol{\beta}_{\parallel} + \boldsymbol{\beta}_{\perp}$, we have

$$\frac{1}{c} \frac{d^2}{dt^2} [(\mathbf{n} \cdot \mathbf{r})\boldsymbol{\beta}_{\perp}] = \frac{1}{c} \frac{d^2}{dt^2} \left(\frac{1}{2}(\mathbf{r} \times \boldsymbol{\beta}) \times \mathbf{n} \right) + \frac{1}{c} \frac{d^2}{dt^2} \left(\frac{1}{2}(\mathbf{n} \cdot \mathbf{r})\boldsymbol{\beta}_{\perp} - \frac{1}{2}(\mathbf{r} \times \boldsymbol{\beta}_{\parallel}) \times \mathbf{n} \right).$$

The last term can be expressed as

$$\begin{aligned} (\mathbf{n} \cdot \mathbf{r})\boldsymbol{\beta}_{\perp} - (\mathbf{r} \times \boldsymbol{\beta}_{\parallel}) \times \mathbf{n} &= \frac{1}{c} \left\{ (\mathbf{n} \cdot \mathbf{r}) \left[\left(\mathbf{n} \times \frac{d\mathbf{r}}{dt} \right) \times \mathbf{n} \right] - \left(\mathbf{n} \cdot \frac{d\mathbf{r}}{dt} \right) [(\mathbf{r} \times \mathbf{n}) \times \mathbf{n}] \right\} \\ &= \frac{1}{c} \left\{ (\mathbf{n} \cdot \mathbf{r}) \left[\left(\mathbf{n} \times \frac{d\mathbf{r}}{dt} \right) \times \mathbf{n} \right] + \left(\mathbf{n} \cdot \frac{d\mathbf{r}}{dt} \right) [(\mathbf{n} \times \mathbf{r}) \times \mathbf{n}] \right\} \\ &= \frac{1}{c} \frac{d}{dt} [(\mathbf{n} \cdot \mathbf{r})[(\mathbf{n} \times \mathbf{r}) \times \mathbf{n}]]. \end{aligned}$$

Putting everything together, $\mathbf{J}$ becomes

$$\mathbf{J} = \frac{e}{c} \int \left(\frac{d^2 \mathbf{r}_{\perp}}{dt^2} + \frac{d^2}{dt^2} \left(\frac{1}{2}(\mathbf{r} \times \boldsymbol{\beta}) \times \mathbf{n} \right) + \frac{1}{c} \frac{d^3}{dt^3} \left[\frac{1}{2}(\mathbf{n} \cdot \mathbf{r})[(\mathbf{n} \times \mathbf{r}) \times \mathbf{n}] \right] \right) e^{i\omega t} dt.$$

For a single particle, if we define

$$\mathbf{p} = e\mathbf{r}, \quad \mathbf{m} = e \frac{\mathbf{r} \times \boldsymbol{\beta}}{2}, \quad Q_{\alpha\beta} = e(3x_{\alpha}x_{\beta} - r^2\delta_{\alpha\beta}), \quad \text{and} \quad \mathbf{Q}(\mathbf{n}) = \sum_{\beta} Q_{\alpha\beta} n_{\beta},$$

we can relate $\mathbf{J}$ to the multipole moments of the particle,

$$\mathbf{J} = \frac{1}{c} \int \left(\frac{d^2 \mathbf{p}_{\perp}(t)}{dt^2} + \frac{d^2 \mathbf{m}(t)}{dt^2} \times \mathbf{n} + \frac{1}{6c} \frac{d^3 \mathbf{Q}_{\perp}(\mathbf{n}, t)}{dt^3} \right) e^{i\omega t} dt.$$

The quadrupole moment requires a bit of explanation. It is easy to show that

$$\mathbf{Q}(\mathbf{n}) = \sum_{\beta} Q_{\alpha\beta} n_{\beta} = e[3(\mathbf{n} \cdot \mathbf{r})\mathbf{r} - r^2\mathbf{n}],$$

and

$$\mathbf{Q}_{\perp}(\mathbf{n}) = (\mathbf{n} \times \mathbf{Q}(\mathbf{n})) \times \mathbf{n} = 3e(\mathbf{n} \cdot \mathbf{r})[(\mathbf{n} \times \mathbf{r}) \times \mathbf{n}],$$

which leads to the expected expression.

It is straightforward to generalize to multiple particle case and we will stop here.